



Crowd Counting Using Video Analytics

Internship Project Report

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Table of Contents

S.no	Title		Page No
	Abstract		1
	Acknowledgements		2
1	Introduction		3
	1.1	Project Background	
	1.2	Problem Statement	
	1.3	Project Objectives	
2	System Design and Technology Stack		4 - 6
	2.1	System Architecture	
	2.2	Technology Stack Rationale	
3	Implementation Details		7 - 8
	3.1	Core Software Modules	
	3.2	Database schema	
4	Results and System Functionality		9 - 13
	4.1	User Authentication	
	4.2	Dashboard Overview	
	4.3	Image & Video Input for Counting	
5	Challenges and Future Scope		14 - 15
	5.1	Challenges	
	5.2	Future Enhancements	
6	Conclusions		16

Abstract

This project presents the development of an “Crowd Monitoring and Safety Management System” designed to automate real-time people detection, counting, tracking, and zone-based safety analysis using advanced computer vision and deep learning techniques. Leveraging the YOLOv8 object detection model, DeepSORT/ByteTrack tracking algorithms, and OpenCV-based frame processing, the system accurately identifies individuals in live camera feeds and assigns persistent IDs to monitor their movement and behavior across predefined zones. An interactive dashboard built using Streamlit/Flask provides live analytics, including people count trends, density heatmaps, and zone occupancy visualizations, enabling users to quickly assess crowd conditions. The system also incorporates intelligent alert mechanisms that trigger notifications when crowd thresholds are exceeded or when individuals remain in danger zones beyond acceptable limits. A secure, role-based authentication model using JWT ensures controlled access for both users and administrators, while the integrated admin panel allows easy management of cameras, monitoring zones, user accounts, alert configurations, and system logs. Additionally, the platform supports exporting detailed analytics through PDF and CSV reports for operational planning and compliance.

By eliminating the limitations of manual crowd monitoring and providing continuous, data-driven insights, this AI-enabled solution enhances safety, improves operational efficiency, and promotes smarter crowd management in environments such as malls, airports, event venues, public buildings, and transportation hubs.

Keywords: YOLO-v8, Real-Time Analytics, Python, Streamlit, Object Tracking, OpenCV, SQLite

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1.Introduction

1.1 Project Background

Crowd monitoring in places like malls, airports, and stadiums is traditionally done through manual observation, which is slow, error-prone, and unable to provide real-time insights. In crowded environments, delays in detecting risks such as overcrowding or restricted-area entry can lead to serious safety issues. With advancements in computer vision and AI, it is now possible to automate this process. Intelligent systems can continuously analyse video feeds, track people accurately, and generate instant alerts, making crowd management faster, more reliable, and proactive.

1.2 Problem Statement

Current manual monitoring processes suffer from several inherent flaws:

- **Inaccuracy:** Human operators cannot accurately count moving crowds over long periods.
- **Latency:** Manual identification of overcrowding often occurs too late to prevent bottlenecks.
- **Lack of Analytics:** Security staff cannot easily generate historical data or density trends needed for future planning.

This project aims to solve these issues by automating the detection process using a computer vision pipeline.

1.3 Project Objectives

The primary objectives of this project are:

- **Automated Detection:** To utilize YOLOv8 for real-time detection and counting of people in video streams and images.
- **Interactive Dashboard:** To develop a user-friendly web interface using Streamlit that displays live analytics, including line charts and density heatmaps.
- **Zone Management:** To enable administrators to define specific areas of interest (zones) and monitor occupancy levels within those boundaries.
- **Reporting:** To facilitate the export of alert logs and analytical summaries in PDF and CSV formats for decision-making.

2. System Design and Technology Stack

2.1 System Architecture

The system follows a modular and streamlined architecture that separates AI-based video processing from the user interface and data management components. The entire application is built as an interactive, full-stack solution using Streamlit for the frontend, YOLOv8 for detection, DeepSORT/ByteTrack for tracking, and SQLite for lightweight database operations. The workflow is designed to ensure smooth real-time performance while maintaining clean communication between modules.

The workflow of the system is as follows:

- **Video Input:** The application accepts video streams from three possible sources—uploaded video files, uploaded images, or a live webcam feed. The input frames are continuously passed into the processing engine.
- **YOLOv8 Person Detection:** Each frame is processed by the YOLOv8 (yolov8n.pt) model, which detects all instances of the "person" class.
- **Object Tracking:** The detected bounding boxes are tracked with unique IDs, enabling continuous person tracking and accurate movement and dwell-time calculation.
- **Streamlit Backend Logic:** It serves as the main controller by managing real-time video display, user interactions, JWT authentication, database communication, dynamic dashboards, and camera/zone configurations.
- **Database (SQLite):** The database stores all essential persistent data, including user accounts, zone boundaries, camera settings, alert history, and count statistics.
- **Interactive Frontend Dashboard:** The frontend displays real-time analytics including live people count, zone occupancy, heatmaps, line charts, and alert notifications. Users can switch between different views, monitor crowd behavior, and export data.
- **Report Generator:** The system can generate downloadable summaries by using Python libraries like Report Lab and Pandas to create PDF reports and CSV logs of crowd activity, alerts, timestamps, and zone-based statistics.



The System Architecture

The system begins with a user registration process where new users create an account. After successful registration, users can log in to access the main application features.

Upon logging in, users are directed to a set of core modules:

- Admin Panel – Allows administrators to manage users, configure system settings, and monitor activities.
- Dashboard – Displays key analytics, charts, and summary metrics for quick insights.
- Video/Image & Live Webcam Module – Enables uploading images, viewing video inputs, and accessing real-time webcam feeds.
- Documentation Section – Provides manuals, guidelines, and reference materials for users.

Finally, the system provides a Logout option to securely end the session.

2.2 Technology Stack Rationale:

Component	Technology	Justification
Web Framework	Streamlit	chosen for its ability to rapidly build data-centric web applications and interactive dashboards using pure Python.
AI Model	YOLOv8	Selected for its state-of-the-art accuracy and speed in real-time object detection.
Video Processing	OpenCV (cv2)	Essential for frame-by-frame manipulation, image annotation, and stream handling.
Database	SQLite	A serverless, lightweight database ideal for managing relational data (Users, Zones, Logs) without complex setup.
Visualization	Plotly Express	Used to generate interactive and responsive charts (bar, line, scatter) for the dashboard.
Authentication	JWT	JSON Web Tokens ensure secure, stateless user authentication and session management.

3.Implementation Details

3.1 Core Software Modules

A. Detection and Tracking Engine

- **Detection:** The system utilizes YOLOv8 to process video frames. It detects individuals in real-time from live feeds or uploaded files with high accuracy.
- **Tracking:** To maintain continuity, DeepSORT is implemented. This algorithm assigns a unique ID to every detected person, tracking their movement across frames to prevent double-counting and visualize their trajectory.

B. Zone-Based Analytics

- **Interactive Configuration:** The project uses **streamlit-drawable-canvas** to allow users or admins to draw specific polygons ("zones") directly on the camera feed.
- **Logic:** The system calculates the center point of each detected person's bounding box. If this point enters a defined zone, the system updates the specific occupancy counter and checks for overcrowding.

C. Interactive Dashboard (Frontend)

- **Framework:** The user interface is built entirely using **Streamlit**, which manages the video display, sidebar controls, and real-time data updates.
- **Visualization: Plotly Express** is used to render responsive charts, including:
 - **Line Charts:** To show total people counts over time (e.g., last 7 days).
 - **Stacked Bar Charts:** To compare occupancy across different zones.
 - **Density Heatmaps:** To visualize the concentration of people within the frame.

D. Data Management and Security (Backend)

- **Database:** A lightweight **SQLite** database is used to store user profiles, camera source configurations, zone coordinates, and historical count records.
- **Authentication:** Security is managed via **JWT (JSON Web Tokens)**, ensuring stateless and secure role-based access (Admin vs. User).

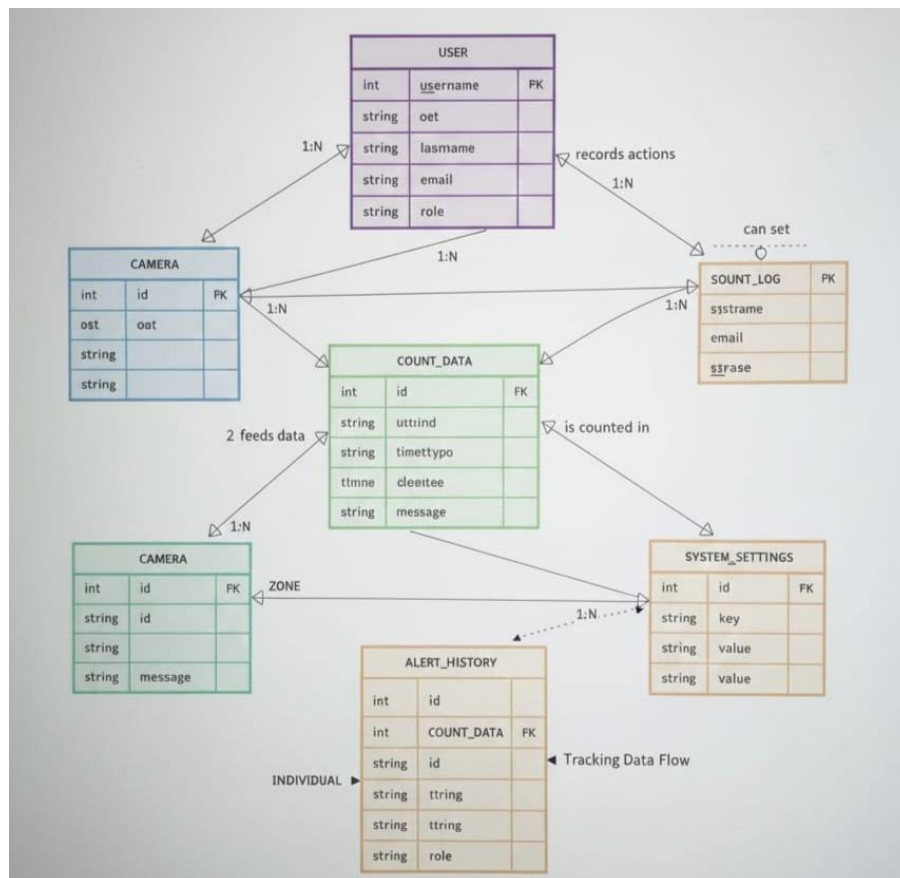
E. Reporting and Admin Tools

The system includes functionality to generate and download summaries.

- **CSV:** Exports raw alert logs and count history.
- **PDF:** Generates formatted analytics reports using **Report Lab** integration.
- **Admin Panel:** A dedicated interface allows administrators to manage users, configure global alert thresholds, and review system logs.

3.2 Database schema

The database organizes the system by storing user details, camera information, and every detection captured by the cameras. All camera outputs go into the COUNT_DATA table, which is checked against system rules to decide if an alert should be created. Any alert is saved in ALERT_HISTORY, while SYSTEM_LOG keeps track of user actions. Zones help group cameras by location, making the data easier to manage. Overall, it shows how camera input becomes count data, how alerts are triggered, and how user activity is recorded.



4.Results and System Functionality

This showcases the final application, demonstrating its core features and functionality through screenshots of the user interface.

4.1 User Authentication

1. Login page

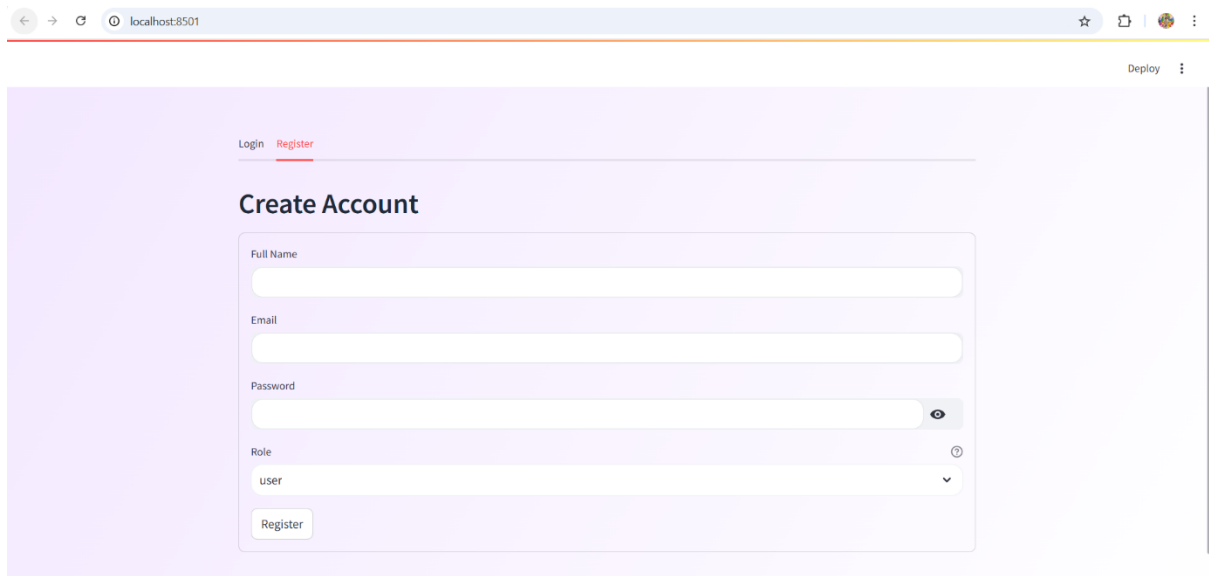
The system provides a secure login interface where users authenticate using email, password, and role-based access.



A screenshot of a web browser displaying the login page of an application. The browser's address bar shows 'localhost:8501'. The page has a light purple background. At the top, there are two links: 'Login' (highlighted in red) and 'Register'. Below these links is a 'Login' section with a form. The form contains three input fields: 'Email', 'Password' (with a toggle for visibility), and 'Role' (a dropdown menu currently set to 'user'). A 'Sign In' button is located at the bottom of the form.

2. Register page

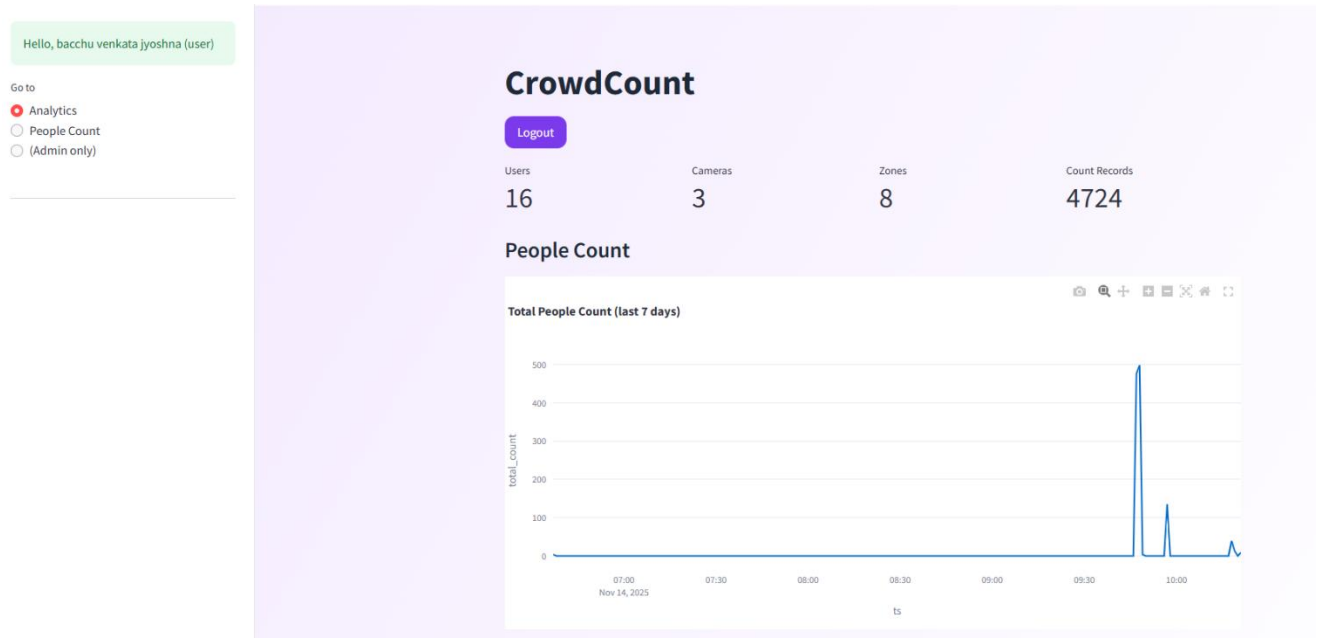
The registration page allows new users to sign up and assign their role as Admin or User prior to logging in.”



A screenshot of a web browser displaying the register page of an application. The browser's address bar shows 'localhost:8501'. The page has a light purple background. At the top, there are two links: 'Login' and 'Register' (highlighted in red). Below these links is a 'Create Account' section with a form. The form contains four input fields: 'Full Name', 'Email', 'Password' (with a toggle for visibility), and 'Role' (a dropdown menu currently set to 'user'). A 'Register' button is located at the bottom of the form.

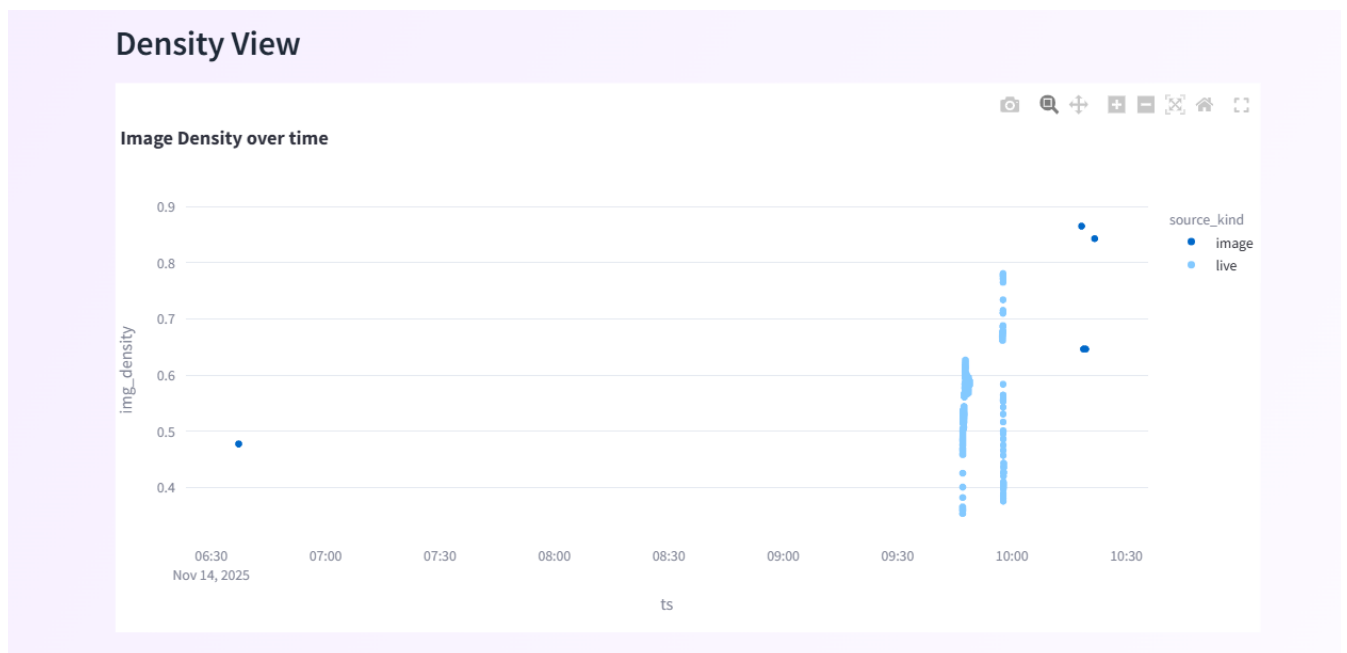
4.2 The Dashboard Overview

Upon logging in, the user is presented with the main dashboard, which serves as the central command center. The dashboard provides real-time analytics, showing total users, cameras, zones, and people count trends over time.



1. Density Visualization:

The density view displays real-time changes in image density extracted from live and uploaded frames.



2. Zone Occupancy Analysis:

The system visualizes zone occupancy with color-coded image/video uploads and also allows users to download the recorded analytics in CSV or PDF format.



4.3 Image & Video Input for Counting

This page allows users to select a camera or upload images to count people and draw monitoring zones.

People Count & Zone Management

Select Camera
#3 Q

	id	name	created_at	type
0	7	Zone from image	2025-11-11T14:15:18.486506	Red
1	6	Zone from image	2025-11-11T14:15:17.408654	Red
2	5	Zone from image	2025-11-06T15:08:50.867816	Red
3	4	Zone from image	2025-11-06T14:36:10.182976	Red

Zone ID: 7 New Label: ☒ Mark as Red Zone

Count People

Image (draw zones) Video File Live (Webcam/IP)

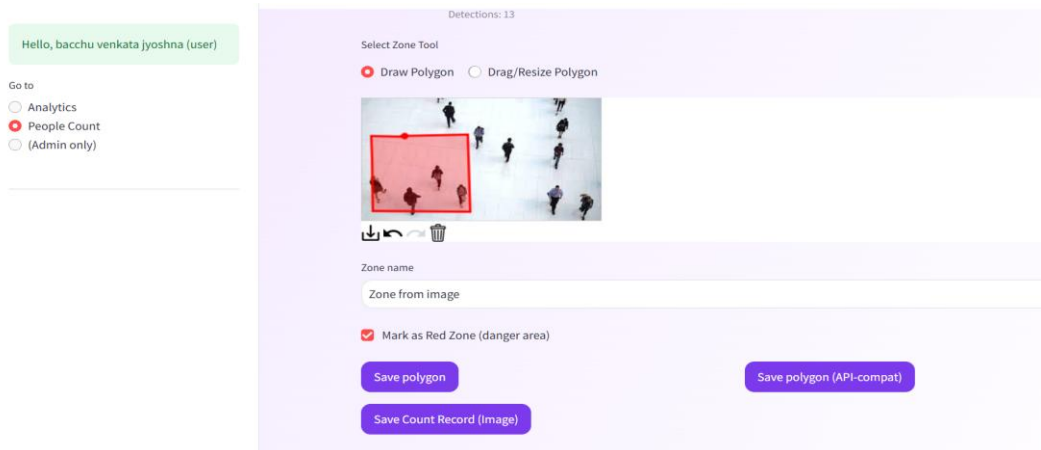
Upload image

Drag and drop file here
Limit: 200MB per file • JPG, JPEG, PNG

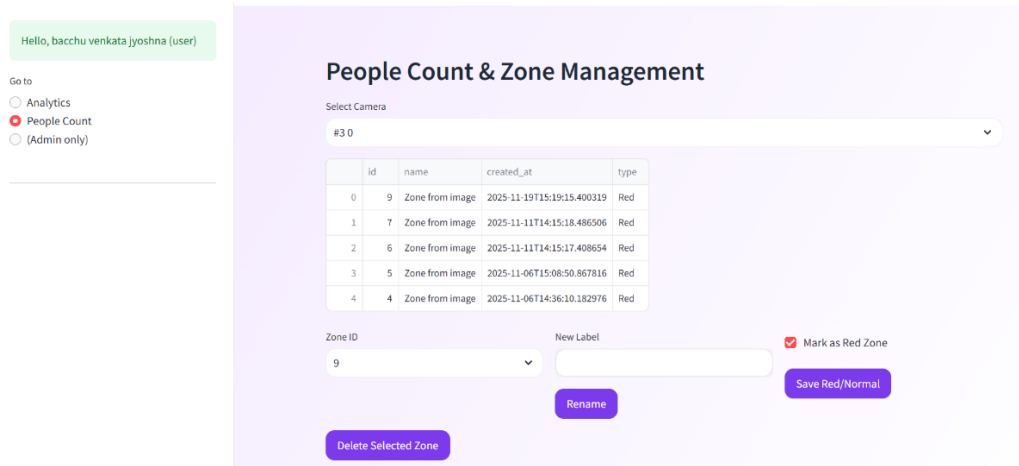
image person 2.jpeg 5.1KB

Detections: 13

Zone Drawing Tool and Deleting/Updating Zones



This interface allows users to draw polygonal zones on the image and save them as monitored or danger zones and giving alerts.



Users can manage zone records by selecting, editing, or removing any saved polygon zone.



Each detected individual is represented with a bounding box, showcasing successful crowd detection.

4.4 Admin Panel

The Admin Panel provides administrators with complete control over the system, allowing them to manage users, cameras, logs, and reports from a centralized interface. The user management table displays detailed records including name, email, role, and account creation time enabling efficient monitoring and administration.

Hello, bacchu venkata jyoshna
(admin)

Go to

☐ Analytics

☐ People Count

☒ Admin

Admin Panel & Analytics

Users

Cameras

Logs

Reports

	id	name	email	role	created_at
0	18	bacchu venkata jyoshna	jyoshna990@gmail.com	admin	2025-11-19T15:51:39.189397
1	17	bacchu venkata jyoshna	jyoshna23@gmail.com	admin	2025-11-19T15:50:25.203590
2	16	bacchu venkata jyoshna	1234@gmail.com	user	2025-11-19T14:41:59.274321
3	15	Sai Chandana	jyoshna10@gmail.com	user	2025-11-14T13:45:49.256827
4	14	Bacchu Venkata Jyoshna	jyoshna12@gmail.com	admin	2025-11-14T10:58:14.471979
5	13	BACCHU VENKATA JYOSHNA	jyoshna78@gmail.com	user	2025-11-14T09:00:40.827243
6	12	bvj	jyoshna3@gmail.com	user	2025-11-14T06:36:29.379982
7	11	BACCHU VENKATA JYOSHNA	jyoshna99@gmail.com	user	2025-11-11T12:24:44.157748
8	10	BACCHU VENKATA JYOSHNA	jyoshna12@gmail.com	user	2025-11-06T15:05:25.172674
9	9	BACCHU VENKATA JYOSHNA	jyoshna11@gmail.com	user	2025-11-06T14:59:26.201168

User Creation & Role Configuration

This form allows the admin to create new users or update existing accounts by entering their name, email, password, and assigned role. The role dropdown lets the admin specify whether the user will function as a general ‘User’ or an ‘Admin’ with elevated permissions.

Create/Update User

Name

Email

Password

Role

user

Save User

5.Challenges and Future Scope

5.1 Challenges

During the development of the Crowd Counting and Analytics System, several challenges were encountered across technical implementation and system integration. Addressing these issues provided valuable opportunities to refine the system's accuracy and performance.

1. Technical Challenges

- **Object Occlusion & Tracking Stability:** In dense crowd scenarios, individuals often crossed paths or were obscured by others, causing the tracker to occasionally swap IDs or lose track of specific persons.
 - **Resolution:** Utilized the ByteTrack/DeepSORT tracking algorithm to maintain persistent IDs and optimized tracking parameters to handle temporary occlusions effectively.

Zone Coordinate Mapping: Accurately mapping the user-drawn zones from the streamlit-drawable-canvas to the actual video frame resolution caused initial misalignment in detection logic.

- **Resolution:** Implemented a coordinate scaling function to translate canvas dimensions to video frame dimensions, ensuring precise boundary checks for the "Danger Zones."

2. Operational & Integration Challenges

- **Real-Time Dashboard Updates:** Integrating the live video processing loop with the Streamlit dashboard interface was challenging, as standard page refreshes would interrupt the video stream.
 - **Resolution:** Leveraged efficient session state management and optimized the frame processing loop to update analytics charts (Plotly) without reloading the entire application logic.
- **Database Latency:** writing high-frequency count logs to the database during live processing initially caused minor delays in the video feed.
 - **Resolution:** utilized SQLite for its lightweight nature and optimized database queries to handle logging asynchronously where possible, preventing bottlenecks in the detection pipeline.

5.2 Future Enhancements

- **Multi-Class Detection:** Add the ability to detect and count not just people, but also vehicles and various objects for richer analytics. Enable classification of detected objects for category-based statistics (e.g., counting cars, bikes, backpacks)
- **Privacy Protection:** Integrate features such as face blurring and masking to comply with privacy regulations and safeguard individual identities. Implement access controls and data encryption to further protect personal information
- **Edge Device Processing:** Optimize the system to run computations directly on cameras or local edge devices, lowering bandwidth requirements and enabling offline analytics.
- **Real-Time Alerting:** Implement instant alerts through mobile apps, SMS, or email to enable immediate response to overcrowding or incidents. Set up configurable alert rules for different zones, times, or conditions.
- **Predictive Analytics:** Leverage historical data and AI models to forecast crowd surges and provide suggestions for proactive management, use predictive algorithms to recommend optimum staffing or resource allocation based on expected occupancy.

6. Conclusions

The project's approach to analytics and real-time alerts empowers administrators to take timely actions, enhancing both safety and operational efficiency. With its modular design and robust functionality, it is ready to evolve with advancements in AI and user requirements, making it a valuable asset for modern smart environments.

Key Achievements:

- **Technology Validation:** The successful integration of **YOLOv8** for high-accuracy object detection with the **Streamlit** dashboard demonstrated a practical, full-stack solution for video analytics.
- **Proactive Safety:** The system achieved its core objective of **enhancing public safety** by implementing precise **zone-based monitoring** and a multi-tier alert mechanism for overcrowding and dwell time breaches.
- **Actionable Data:** Through the secure, role-based admin panel, the system provides administrators with **real-time insights** and **detailed exportable reports** (PDF/CSV), supporting data-driven decisions regarding resource allocation and operational planning.

Overall, the project not only improves crowd management and safety but also provides the flexibility needed to adapt to complex real-world scenarios. With planned enhancements—such as multi-class detection, predictive analytics, and IoT integration—the system is well-positioned to evolve into a highly **scalable** and **customizable** solution for next-generation smart monitoring in various public environments.